

Problem Statement

"RetailMart, a fast-growing e-commerce company, is struggling to answer basic business questions. Their data is scattered across raw CSV files — orders, customers, products, payments — sitting on different systems with no central place to analyze it. The business team cannot answer: Which products are trending? Which customers are about to churn? What was last month's revenue? The data engineering team has been asked to build a centralized analytics platform that ingests raw data, cleans it, maintains product and customer history.

Tech stack used:

Python Basics	Load CSVs, count orders
Pandas	Clean orders, delivery days
SQL SELECT/Keys	Order-customer joins
WHERE + Indexes	Filter by order status
GROUP BY	Monthly revenue
JOINS	Customer 360 (Unified view)
CASE	Customer segments
Subqueries	Above-average spenders
CTEs	Funnel analysis
Window Functions	Product rank by category
PySpark	Revenue at scale
Delta Lake	Product catalogue SCD2
Medallion	Bronze/Silver/Gold orders